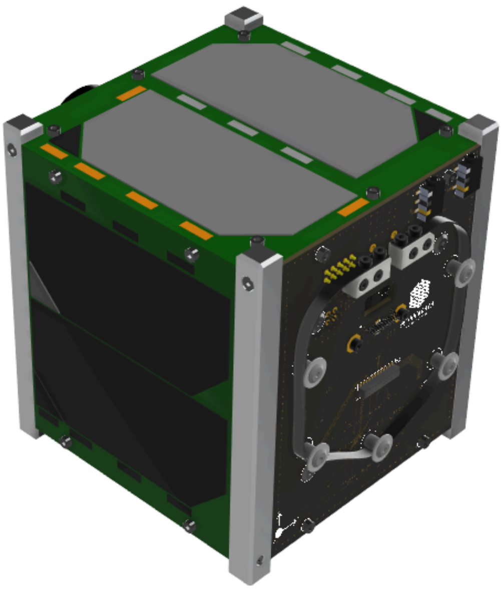
ANTARIKCHYA
अन्तरिक्ष प्रतिष्ठान नेपाल

slippers2sat-2

Transmission Protocol for Slippers2Sat-2



<u>Prepared By</u>

PRAGATI BASNET
SAJAN DUWAL
(Communication System - Slippers2Sat-2)

Slippers2Sat-2 (S2S-2) features a single on-board transceiver capable of alternating between Continuous Wave (CW) and Gaussian Minimum Shift Keying (GMSK) modulation schemes. S2S-2 utilizes a CW Morse code beacon for Space-to-Earth transmissions. For primary data retrieval and mission control, all telemetry downlinks and telecommand uplinks utilize a GMSK modulation scheme.

Communication Timing Diagram

Normal Mode	CW Beacon	Satellite Operation and Digipeater	CW Beacon	Satellite Operation and Digipeater
		90 sec		90 sec

Safe Mode	CW Beacon	Satellite Operation and Digipeater	Callsign and Name of Satellite	Satellite Operation and Digipeater
		NO TRX		NO TRX

Critical Mode	No CW Beacon	Satellite Operation and Digipeater	No CW Beacon	Satellite Operation and Digipeater
		NO TRX		NO TRX

The satellite relies on a time-multiplexed transmission schedule divided into three operational modes based on battery state-of-health and mission commands:

1. Normal Mode: This is the standard 24/7 operational state. The satellite alternates between transmitting the CW Morse code beacon and a 90-second transmission window dedicated to standard Satellite Operations (housekeeping telemetry, ADCS logs, and payload mission data) alongside the active software Digipeater service.
2. Safe Mode: If the battery voltage drops below a safe threshold, the satellite automatically enters Safe Mode. In this state, the satellite continues to transmit its CW Morse code tracking beacon periodically for tracking and basic health monitoring, but it completely cuts off the 90-second transmission window (NO TRX). All main satellite operations and digipeater transmissions are paused to conserve power.
3. Critical Mode: In a critical power emergency, the satellite deactivates all automatic RF transmissions. In this state, there is No CW Beacon and No TRX during the satellite operation/digipeater window. The spacecraft remains completely silent, utilizing all harvested solar energy to charge the batteries, and will only wake up or re-enable transmissions upon a specific battery voltage threshold.

The satellite can transmit the data in every 1 second interval to the ground station. The antenna used in Satellite is a dipole antenna in UHF Amateur Band (435.313MHZ and 437.375MHz).

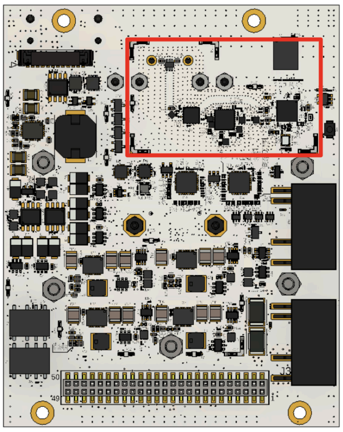


Figure 1. CUBUS_V3 [Showing Communication System]

CW Transmission

Slippers2Sat transmits CW Beacon with callsign and general housekeeping data of the satellite. The Beacon signal can be used to detect the presence of the satellite. It accommodates parameters of the electrical power system, timestamp, antenna deployment status, OBC Reset count, Satellite operation mode, etc, which is transmitted in the regular interval of 90 seconds.

CW Format

- Call Sign
- Satellite Name
- Primary data of Electrical power system
- Primary data of OBC

CW Beacon

- Frequency: 436.375 MHz
- Modulation: CW Morse code
- Data rate: 20 wpm

The received CW will be in HEX format. To convert it to DEC later decoder will be published in https://s2s2.antarikchya.org.np/.

GMSK Transmission

Slippers2Sat-2 uses GMSK modulation scheme to control, retrieve the payload and BUS of the satellite. The configuration for the GFSK modulation scheme are as follows;

Emission Designators: 8K50F1D
Deviation: 12KHz
Baud Rate: 4800 bps
Packet Structure: AX.25 Packet Protocol and G3RUH encoding

For GMSKTelemetry

- **Telecommand**
  - Main Satellite Command: 13 bytes
  - AX.25 Packet Protocol
- **Mission Data/ Telemetry**
  - AX.25 Packet Protocol
  - Upto 100 bytes

SN	Type	Length	Data Type	Value	Remark
1	Flag	1	HEX	7E	
2	Call Sign	14	ASCII		
3	Control	1	HEX	03	
4	PID	1	HEX	F0	
5	Packet type	1	HEX		
6	Packet Count	1	HEX		
5	Information	1-100	HEX		
6	CRC	2	HEX		Calculate the sum from callsign to information byte
7	Flag	1	HEX	7E	

NOTE: The document will be updated before launch. The decoder will be available after complete testing.